Varun Menon

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EDUCATION

Harvard University **Sep 2023 – Present**

Doctoral Candidate (Ph.D.), Theoretical Physics (in progress) | Ashford Graduate Fellow

A.M. Quantum Science and Engineering (Sep 2025)

University of California, Berkeley **Aug 2018 – May 2022**

B.A. Physics (Honors), B.A. Computer Science (Honors), B.A. Applied Mathematics | High Distinction | GPA: 3.97

RESEARCH & WORK EXPERIENCE

Lukin Group, Harvard University – Graduate Student Researcher, Cambridge, MA **Sep 2023 – Present**

Research: Quantum computation & error-correction | Advisor: Mikhail Lukin, Professor of Physics

- Developing efficient fault-tolerant schemes for Clifford and non-Clifford logic on quantum low-density parity-check (qLDPC) codes.
- Designing quantum algorithms to probe spectral properties of interacting many-body systems in condensed matter and quantum chemistry applications.

Google Research (Google Quantum AI) – Student Researcher, Los Angeles, CA **May 2026 – Aug 2026**

- Constructed efficient block-encoding algorithms for Hamiltonian simulation of long-range interacting systems.
- Developed a shadow Hamiltonian simulation quantum algorithm in the Heisenberg picture with super-quartic speedups over classical algorithms for certain interacting fermionic systems.

IBM Research – Quantum Research Scientist Intern, Yorktown Heights, NY **May 2024 – Aug 2024**

- Investigated many-body simulation applications with potential quantum advantage in anomalous Floquet topological phases via large-scale tensor network and sparse Pauli-propagation simulations, and analytical arguments.

Q-CTRL – Quantum Computing Research Scientist, Los Angeles, CA **Jun 2022 – Aug 2023**

- Developed machine-learning-based error mitigation to improve qubit coherence times and gate fidelities on IBM, Rigetti, and OQC superconducting quantum computers, including reinforcement learning control for single- and two-qubit gates, crosstalk-suppressing dynamical decoupling, and readout error mitigation. Achieved over 1300X and 30X accuracy gains for 15-qubit Bernstein-Vazirani and 7-qubit Quantum Fourier Transform algorithms, respectively, on superconducting hardware.

Moore Group, UC Berkeley – Undergraduate & Post-Baccalaureate Researcher, Berkeley, CA **Aug 2021 – Dec 2022**

Research: Many-body quantum theory | Advisor: Joel Moore, Professor of Physics

- Developed matrix product state (MPS) and projected entangled pair state (PEPS) tensor network simulations of the dynamic structure factor in the 1D and 2D Heisenberg models; collaborated with neutron scattering groups at Oak Ridge and Los Alamos National Laboratories to detect entanglement in Heisenberg materials.

SELECT PUBLICATIONS

- *Single-shot magic state factories for quantum LDPC codes*; **V. Menon**, R. Mehta et al.; arXiv:2610.10731; 2026
- *The practical cost of magic state cultivation*; R. Mehta, **V. Menon** et al.; arXiv:2610.06828; 2026
- *Efficient logic with ultra-high-rate quantum codes*; N. Maskara, R. Mehta, Z. He, **V. Menon** et al.; arXiv:2610.06749; 2026
- *Syndrome extraction circuits with near-optimal depths for practical quantum error correcting code families*; D. B. Tan, J.P.B. Ataides, **V. Menon** et al.; Proc. 63rd ACM/IEEE Design Automation Conference (DAC '26); 2026
- *Efficient fidelity simulation of high-rate magic distillation circuits*; X. Xiao, D. Hangleiter, J.P.B. Ataides, R. Mehta, **V. Menon** et al.; arXiv:2610.03605; 2026
- *Memory-efficient quantum simulation via operator-space truncation*; with Google Quantum AI team; manuscript in preparation
- *Applications of shadow Hamiltonian simulation to equilibrium dynamics*; with Google Quantum AI team; manuscript in preparation
- *Realizing anomalous Floquet phases and time-crystalline transport in an interacting disordered spin-system on superconducting quantum computers*; **V. Menon** et al. (with IBM Quantum); manuscript in preparation
- *Observation of multipartite spin entanglement in a cuprate chain*; S. F. R. TenHuisen, Z. Shen, V. Bhartiya, **V. Menon** et al.; arXiv:2609.08134; 2026
- *Magic tricycles: Efficient magic-state generation with finite block-length quantum LDPC codes*; **V. Menon**, J.P.B. Ataides et al.; Phys. Rev. X 16, 021014; 2026
- *Non-Clifford gates between stabilizer codes via non-Abelian topological order*; R. Sajith, Z. Song, B. Roberts, **V. Menon** et al.;

PRX Quantum 7, 010361; 2026

- *Symmetries, correlation functions, and entanglement of general quantum Motzkin spin-chains*; **V. Menon**, A. Gu, R. Movassagh; arXiv:2408.16070; 2024
- *Multipartite entanglement in the one-dimensional spin-½ Heisenberg antiferromagnet*; **V. Menon** et al.; Phys. Rev. B 107, 054422 (Editor's Suggestion); 2023

PRESENTATIONS & CONFERENCES

- *Efficient magic-state generation with tricycle quantum LDPC codes*; **V. Menon**, J.P.B. Ataides et al.; Quantum Error Correction (QEC), 2026; Talk
- *Magic-state generation with finite-length quantum LDPC codes*; **V. Menon**, J.P.B. Ataides et al.; Theory of Quantum Computation (TQC), 2026; Talk
- *Simulability and exact results in a class of highly entangled spin chains*; **V. Menon**, A. Gu, R. Movassagh; Quantum Information Processing (QIP), 2025; Talk
- *Quantum Motzkin spin-chains: exact results and classical simulation*; **V. Menon**, A. Gu; Google Quantum AI Tensor Network Seminar, 2024; Invited Talk
- *Finite temperature critical scaling of multipartite entanglement in the 1D spin-½ antiferromagnetic Heisenberg chain*; **V. Menon** et al.; APS March Meeting, 2023; Talk
- *Improving algorithmic performance of tunable superconducting qubits using deterministic error mitigation*; **V. Menon** et al.; APS March Meeting, 2023; Talk
- *Using machine learning for autonomous selection of optimal qubit layouts on quantum devices*; A. Barbosa, Y. Baum, P. Mundada, G. Hartnett, **V. Menon**; APS March Meeting, 2023; Talk

AWARDS & HONORS

- Theodore H. Ashford Graduate Fellowship, Harvard University, Fall 2023 – Spring 2028
- High Distinction in General Scholarship, UC Berkeley, May 2022
- Highest Honors in Physics (thesis honors), UC Berkeley, May 2022
- UC Berkeley EECS Honors scholar (non-thesis honors), UC Berkeley, May 2022
- Berkeley Physics Undergraduate Research Scholar (BPURS), Fall 2021, Spring 2022
- Lawrence Berkeley National Laboratory Quantum Systems Accelerators Summer Research Fellowship, Summer 2021
- The Kavli Nanoscience Institute, Caltech, SURF-the-Wave undergraduate research Fellowship – one of seven recipients, Summer 2019
- UC Berkeley Edward M. Kraft Award for Academic Excellence, Spring 2019

TEACHING

- Course Tutor: Quantum Mechanics, UC Berkeley, Fall 2020
- Course Tutor: Statistical and Thermal Physics, UC Berkeley, Spring 2022
- Teaching Assistant: Designing Information Devices and Systems I, UC Berkeley, Spring 2020, Fall 2020, Spring 2021

COMMUNITY ENGAGEMENT

Volunteer, Broken Tail Animal Rescue, Cambridge, MA **May 2025 – Present**

Caring for and socializing local rescued and stray animals.

Mentor, Computer Science Mentors (CSM), UC Berkeley **Jan 2019 – May 2019**

Organized peer mentoring sections and exam review sessions for multiple Electrical Engineering and Computer Science Courses.

Tournament Director, Debate Society of Berkeley, UC Berkeley **Jan 2019 – Dec 2019**

Organized a national charity tournament for high school students competing in parliamentary style debate, hosted at UC Berkeley through the Debate Society of Berkeley.